



STEP Interface Functional Specification

Temasek Polytechnic – Library Systems

Version 1.0



Document Version

Version	Effective Date	Summary of Changes	Created/ Updated By
0.1	14/08/2023	Initial version of TP library systems IFS	AvePoint
0.2	22/08/2023	Add one more course type – WSQ	AvePoint
1.0	25/08/2023	Accepted interface specification version by TP and AvePoint	AvePoint



Document Acceptance Form

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1. Introduction

1.1 Purpose of this document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirements and detailed technical specification between the SaaS Skills Training Enhancement Portal (STEP) and Temasek Polytechnic Library Systems to interface students data for the creation of Continuing Education & Training (CET) students' library accounts and for facial recognition gantry access into the library.

STEP shall identify students' details based on the CIS Student ID, enrolment information, course end date and personal information, such as name and email address, to generate corresponding CSV and XML files. STEP shall generate and deposit the CSV and XML files encrypted by PGP protocol in the STEP SFTP server accordingly. Temasek Polytechnic can create student library account and grant gantry access based on the transferred files.

1.2 Intended Audience

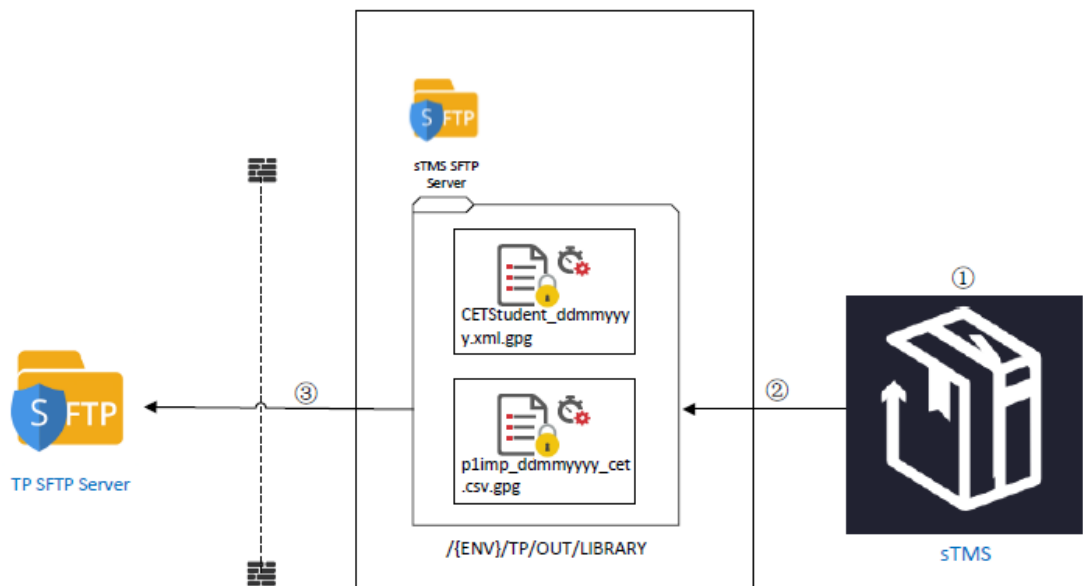
This technical document is used for system communication. Therefore, the intended audiences are the technical stakeholders of TP Library System and STEP.

2. Technical Overview

2.1 System Flow

There are 3 roles involved in the interface diagram.

- **TP SFTP Server:** Temasek Polytechnic SFTP Server.
- **STEP SFTP Server:** A SFTP server hosted in Azure that stores encrypted file used for data exchange.
- **STEP:** Skills Training Enhancement Portal



SN	Description
1	STEP generates the XML and CSV files based on the student's enrollment status, the student only enrolled in FQ will be exported. Please refer to TP FQ & Non-FQ list here .
2	STEP configures daily timer job push to STEP SFTP server after encryption with PGP protocol.
3	TP pulls files from STEP SFTP server for downstream system to consume data.

2.2 Interface Design Summary

Output Interface File

Interface File Name	CETStudent_ddmmyyyy.xml p1imp_ddmmyyyy_cet.csv
Data Source System	Skills Training Enhancement Portal
Data Destination System	Temasek Polytechnic Library Systems
Mode of Interface	CSV file with “,” as delimiter; XML file
Method	SFTP
File Path	/{ENV}/TP/OUT/LIBRARY ENV: UAT, PROD
Encryption Method	PGP
Data Syncing Mode	full load data sync
Frequency	Daily
Timing	Configurable
File Retention Policy	Files are retained for a maximum of 30 days in the STEP SFTP server, after this period the files will be automatically deleted
Possible Exception and Exception Handling	1. File loss or transfer failure due to STEP internal exception. E.g. Timer service down, etc. Exception handling steps: - STEP support team shall be informed by email which sent by STEP of the transfer failure. - AP team checks the relevant logs to troubleshoot the cause of the failure. - If the issue is due to the transfer file could not be found in STEP SFTP server: - STEP support team will manually rerun the timer job to trigger file generation and deposit the file in STEP SFTP server. 2. Data error. Exception handling steps: - STEP support team shall be informed by email which sent by STEP of the file generation failure. - AP team checks the relevant logs and file to troubleshoot the cause of the failure (data error types are listed below). - Field is mandatory but missing - Field failed validation - AP redeposits the correct file in SFTP.

2.3 Business Rule

1. The active student information would be sent if enrolled in FQ & Non-FQ courses. Refer to course type belongs to FQ & Non-FQ.

Course Type	Code	Remark
Academic Full Qualification	FQ	FQ
Academic Certificate	MC	FQ
Academic Modular	SBMC	FQ
Work-Study Certificates Leading To Part-Time Diploma	WSC2PTD	FQ
Work-Study Post-Diploma Specialist Diploma	WSPSD	FQ
Work-Study Post-Diploma Leading To Specialist Diploma	WSP2SD	FQ
Work-Study Post-Diploma Advanced Diploma	WSPAD	FQ
Work-Study Post-Diploma Diploma Conversion	WSDC	FQ
Workforce Skills Qualification	WSQ	Non-FQ

2. STEP will send the following two files containing student information to TP SFTP Server:
 - a. Students attending FQ & Non-FQ would be recorded in CETStudent_ddmmyyyy.xml for library account creation.
 - b. Students attending FQ & Non-FQ would be recorded in p1imp_ddmmyyyy_cet.csv for library facial recognition photo.
3. Mapping of STEP student enrollment status to include in data files, including the logic of start date, expiry date, and school code.

STEP student status	Processing single course	Processing multiple courses
Enrolled, Ready to complete, Ready to graduate,	Include in data file. [start date]: Today [expiry date]: courseIntake's last semester's end date	Include in data file. [start date]: Today [expiry date]: end date of course with longest expiry date [school]: school code of course with longest expiry date
Withdrawn, Deferred	Include in data file after withdrawn/deferred + 7 days [start date]: Yesterday [expiry date]: Yesterday	Include in data file. [start date]: Today [expiry date]: expiry date of remaining enrolled course with longest expirydate [school]: school code of remaining course with longest expirydate
Transferred to new FQ intake	Include in data file. [start date]: Today [expiry date]: new courseIntake's last semester's end date	Include in data file. [start date]: Today [expiry date]: end date of course with longest expiry date [school]: school code of course with longest expiry date
Graduated, Completed	Include in data file until course completion + 7 days [start date]: Yesterday [expiry date]: Yesterday	Include in data file [start date]: Today [expiry date]: expiry date of remaining enrolled course with longest expirydate

		[school]: school code of remaining course with longest expirydate
Suspended	Include in data file after suspended + 7 days [start date]: Yesterday [expiry date]: Yesterday	Include in data file after suspended + 7 days [start date]: Yesterday [expiry date]: Yesterday
Pending enrolment	Do not include in data file	Include in data file. [start date]: Today [expiry date]: end date of course with longest expiry date [school]: school code of course with longest expiry date

2.4 Interface File Generation Rule

Flow to Generate Library Data

SN	Description
1	STEP identifies the student's enrollment status daily. Once the student is enrolled, STEP shall locate the corresponding student information and course information. Refer to 3. Interface Fields for the fields to be sent.
2	STEP generates CSV and XML files to consolidate the full-load data of students.
3	STEP deposits the CSV and XML files in SFTP server accordingly.

3. Interface Fields

Student information for library account creation:

TP Field	STEP Name	Description	Data Type	Remarks
record_type	-	-	String(\$nvarchar(6))	Default value: PUBLIC
preferred_language	-	-	String(\$nvarchar(2))	Default value: en
account_type	-	-	String(\$nvarchar(8))	Default value: EXTERNAL
external_id	-	-	String(\$nvarchar(8))	Default value: SIS_temp
status	-	-	String(\$nvarchar(6))	Default value: ACTIVE
primary_id	Student ID	CIS common ID with domain name Whole string in uppercase.	String(\$nvarchar(50))	Sample value: 6159137P@polite.edu.sg
last_name	Student Name	Student Full Name Whole string in uppercase.	String(\$nvarchar(50))	Sample: John Smith
user_group	-	-	String(\$nvarchar(2))	Default value: 10
expiry_date	Course end date	Refer to business rule for detailed information	String(\$nvarchar(10))	Format: YYYY-MM-DD
purge_date	Course end date	Same as expiry_date	String(\$nvarchar(10))	Format: YYYY-MM-DD
email_type	-	-	String(\$nvarchar(10))	Default value: personal
email_address	Student personal email address	Student personal email address	String(\$nvarchar(50))	Smith.john@gmail.com

user_identifier-id_type	-	-	String(\$nvarchar(2))	Default value: 03
user_identifier-value	Student ID	CIS common ID Whole string in uppercase.	String(\$nvarchar(50))	Sample value: 6159137P
<category_type>Patron Category</category_type>	-	-	-	Default value: CET
<category_type>School/Dept</category_type>	School code	Refer to business rule for detailed information Whole string in uppercase.	String(\$nvarchar(10))	Sample value: ASC; BUS

Student information for library facial recognition photo:

TP Field	STEP Name	Description	Data Type	Remarks
Staff No	Student common ID	Student common ID Whole string in uppercase.	String(\$nvarchar(10))	Sample value: 2012345A
Staff Name	Student common ID	Student common ID Whole string in uppercase.	String(\$nvarchar(10))	Sample value: 2012345A
Card Number	130+Student common ID	CET number starts with 130+common ID Numeric only. Drop the alphabet	String(\$nvarchar(10))	Sample value: 1302012345
Card Start Date	Today's date	Today's date, but start date can not later than end data, refer to business rule	String(\$nvarchar(11))	Format: DD-MMM-YYYY
Card End Date	Course end date	Refer to business rule for detailed information	String(\$nvarchar(11))	Format: DD-MMM-YYYY



PushToFR	-	-	-	default value: 1
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4. Sample Files

